Lindsey (Lanxin) Ma

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Dear Black Canyon Consulting Hiring Team,

I am applying for the Junior Data Scientist position. I hold an M.S. in Statistics from Columbia University (May 2026), and the role's mix of statistical analysis in Python, careful evaluation of AI/ML methods, and work with language models matches what I have been doing.

As a Research Assistant at Columbia's Center for Theoretical Neuroscience, I worked with noisy scientific data: I designed a self-supervised denoising method for two-photon calcium imaging that reduced residual energy by up to 20%, built synthetic benchmarks to measure signal recovery across noise levels, and packaged the lab's research code with documentation for wet-lab researchers. That last part taught me to explain methods clearly to people who are not statisticians.

At Arborphy AI I build and evaluate machine learning models, and I put a lot of weight on whether an evaluation can be trusted. I designed a leakage-aware protocol with individual-disjoint splits and multi-seed runs, and it showed that an apparent gain from a more complex model did not reliably beat a simpler baseline. I would bring the same skepticism to judging AI/ML methods for appropriateness, bias, and validity in clinical research.

I have also worked directly with language models: I reproduced a circuit-tracing analysis of hallucination on Qwen3 and built interpretability tooling across all 28 transformer layers, and I integrated a locally hosted language model to map voice commands to robot actions.

I would welcome the chance to support BCC's research data work. Thank you for your consideration.

Sincerely,
Lindsey Ma